

Deleting a Memory and Running It Again

An instrument for checking what a model says about its own context

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Abstract

Ask a language model which part of its context made it answer the way it did, and it will tell you. Checking the answer means running the counterfactual — rebuilding the request without that item and asking again. Most stacks flatten the context into a string, so its items stop being addressable and it is rarely done. I built a runtime in which a session's context is addressable state, so a session can be forked, one named item removed, and the copy asked the same question again. **This paper is about the apparatus** — what it takes to make a measurement like that honest — and what it found on six models from six labs, twice. The registered hypotheses failed; the second collection was taken on a corrected instrument against four thresholds registered and timestamped beforehand, and all four held.

The confirmed result is that these models' false positives have a shape. Across both collections, 0 of 135 planted irrelevant notes were ever claimed to matter, 1 of 199 notes without figures was, and 62 of the other 63 fell on notes belonging to the question's arithmetic that did not decide it; the one exception is task-central and carries no quantity at all. Models over-weight what looks like a reasoning step, not what looks numerical.

Models are also reasonably good at saying *what* their answer rests on — 72–96% correct then 72–85%, above their own majority baseline in four of six models and then five of six — but most items are inert, so accuracy flatters them. Recall and precision run 43–100% and 36–87%, with `solar-pro4` worst on both, and most of the recall spread is sampling rather than skill: restricted to items where both arms of three copies were unanimous, pooled recall goes from 77% and 66% to 92% and 89% while precision barely moves. Two further measurements are weaker and reported as such — no own-context advantage was detected, at five items per arm per model, which is a failure to detect rather than a null; and on a five-rung repair ladder *telling* a model one of its notes is false helped every model that could be measured, while *then asking it to put the context right* moved one measurement of seven past the registered ceiling and left the others between −20 and +13 points. Every question and answer from both collections is saved, and every number here recomputes from that record.

1. Introduction

This did not start as research. I was building a runtime for language-model agents in which the context — the pile of messages, notes and tool results that goes into each request — is proper state rather than a string that gets concatenated. Every item has an identity and a lifecycle. You can snapshot a session, change one thing, and resume it.

Once that existed, an experiment became possible that is otherwise very hard to run. If you want to know whether a model's answer really depended on a particular note, you can take the note out and ask again. Two copies of the same session, identical except for one item, both asked the same question. If the answers differ, the item was load-bearing. If they do not, it was not.

That is worth pausing on, though not for the reason I first wrote down. In fifty years of work on human self-report — starting with Nisbett and Wilson's *Telling More Than We Can Know* (1977) — the standing problem is that you cannot rerun a person with one memory deleted, so a stated reason can be shown *implausible* and never *wrong*. A

language model can be rerun. But prompt-level ablation and replay are not new either; §8 names the closest published work, which makes the same move first. What the runtime adds is only that the item is addressable rather than a substring, so the ablation is a named operation on state — cheap in bulk, and recorded item by item so somebody else can check it. That is convenience and auditability, not new capability.

So I used it to check a simple thing: when a model explains what its answer depends on, is it right? I expected it not to be. I was wrong, and the ways in which I was wrong are the useful part.

But the measurements are not the part worth your time. Six models will be deprecated inside a year and their numbers with them. What survives is the apparatus, because **running the counterfactual is the easy half**. The hard half is everything that stops the result from being an artefact: what to baseline against, what to refuse to score, how to tell your own failure from the subject's, and how to know whether two runs were asked the same question at all. Every one of those is a decision I got wrong at least once, and §5 is the list.

This is not a peer-reviewed study. I am a Rust engineer rather than a researcher, nobody who reads this literature regularly has checked it, and it should be read as a report on an instrument and what it measured. §7 says who did what, including the parts a language model did; Appendix B is the dated log of everything that changed after the freeze, including two withdrawn results.

2. Method

2.1 The setup

Each trial plants a small dossier of notes in a fresh session — a set of records about, say, three warehouses — and asks a question the notes answer, such as which one runs out of space first. The model answers.

Then, for each note, two things happen:

- **The model is asked:** "if this note were removed, would you answer differently?"
- **The harness finds out:** it forks the session, removes that note, and asks the question again. Three copies, to check they agree with each other, and three more under the unablated control.

The copies never see the exchange in which the model already answered, so they are not being asked to remember, only to work it out again. The model's claim is then scored against what the copies actually did. Neither side gets special treatment: both are the same model, on the same context, minus one item.

2.2 The materials

Six dossiers, each with nine notes and a three-way question. Every dossier contains:

- notes that carry the arithmetic of the question (capacities, rates, stock levels),
- one note that decides the answer, usually a correction buried in prose,
- inert notes with no bearing on the question at all,
- and **two "red herrings"** — notes stating a plausible figure for each of the three options on a dimension that has nothing to do with the question. Road distances to a regional office. Rateable values. Deed reference numbers.

The red herrings exist because of a defect I found in my own material partway through. Before they were added, *every single inert note in the whole set had three digits or fewer*. So "contains numbers" and "matters" were tangled together, and a model that simply guessed "notes with figures in them are important" would have scored

well for the wrong reason. With one red herring per dossier that shortcut still scored 0.83 against the truth — better than the models themselves managed — so I added a second. Two brings it to 0.74, and a test in the repository holds it there.

2.3 What is scored

The main measurement is deliberately narrow. It looks **only at notes whose removal did not change the answer** — where the treated copies and the control copies came out the same. For all of those, the correct claim is "no, this doesn't matter". So among items that all do nothing, does the model claim the numeric ones matter more often than the plain ones? What "came out the same" means exactly, and what a stricter reading of it costs, is §2.4.

Restricting it that way is the whole design. Across the material as a whole, notes with figures really *are* more likely to matter, because that is where the arithmetic lives. An unrestricted comparison would reward a model for a lucky prior. Inside a group of items that are all scored as inert, there is nothing left to be right about, so any gap between the halves cannot be knowledge.

What is scored, throughout, is a prediction of the **ablation outcome**: whether removing the item changes what the model says. That is not the same as whether the item influenced the original computation. An item can be read, reasoned over and still be redundant, and its removal will change nothing — the apparatus will call it inert, correctly, for the question it is asking. Every claim in this paper is about that question and no larger one.

2.4 The gate that was registered and not applied

The harness checks its own preconditions before printing anything — that removing the decisive item moves the answer, that the control copies agree, that the subject answered the dossier as its notes support, and in the ladder that the falsehood fooled it — and prints every failure beside the numbers as an `unmet:` line. §5 is what those checks are for. One of them was registered as a gate and never applied, and that needs stating here because it decides what §4.3's numbers are. §8.2 of the preregistration reads:

If the control copies disagree with each other, the condition is reported with its instability figure and excluded from the primary analysis.

The instrument reports the instability figure and excludes nothing. It compares the *unique plurality* of the readable copies on each side: a control that tied has no answer and drops out by construction, but one that went two-one still yields a plurality and the condition stays in. §4.3 is therefore plurality-scored rather than gated, and is kept that way because that is what was run.

How often the gate would have fired. The three control copies were unanimous on 30 of the 36 dossiers in each collection; the other six went two-one, except one dossier in the second collection where all three differed. That is 54 of 324 ablations on a control that did not agree, 17% both times, mean instability 0.056 then 0.065. Those 54 are not the 35 and 33 the gate drops, and the difference is not a discrepancy: the gate can only drop items the endpoint contains and the endpoint is inert items alone, so 35 and 33 of the 54 were inert and dropped, while 18 and 21 were load-bearing and never entered it, plus one first-collection item never measured.

What it costs is small. Both collections below, first then second; `herrings` and `off-pivot` partition `numeric` rather than adding to it:

reading	items dropped	numeric	plain	herrings	off-pivot	discrimination
as published	—	32/140, 31/136	0/101, 1/98	0/69, 0/66	32/32, 31/31	6/6, 6/6
§8.2 as registered	35, 33	26/119, 25/117	0/87, 0/84	0/59, 0/58	26/26, 25/25	6/6, 6/6
both arms unanimous	57, 55	21/103, 19/99	0/81, 0/80	0/51, 0/52	21/21, 19/19	5/6, 6/6

The registered gate moves no conclusion: all four replication thresholds hold and the direction is unanimous both times. It does remove the one plain-note false positive, which sat on a material whose control had not agreed, so under §8.2 the plain count is 0 of 171 rather than 1 of 199. The third row asks the same of the treated batch, which nothing registered, so it is a sensitivity analysis chosen after the fact; it is the only row where something gives, and what gives is the unanimity of direction in the first collection, 6 of 6 becoming 5 of 6.

One refusal belongs here rather than in §5, because it defines what is being compared. A session that has committed to an answer and is then copied would be reading its own commitment. Both arms are blinded to the exchange in which the subject already answered, so what differs between them is the item and nothing else.

3. What was fixed in advance

The hypotheses, the numeric predictions, the exclusion rules and a nine-row table of "if the results look like *this*, do *that*" were written down before collection and committed to a separate public branch. That matters more than usual here, because I generated the hypotheses by looking at early exploratory runs — a legitimate way to *find* a hypothesis and a terrible way to *test* one, so the fix is to register it and collect fresh data. Which is what happened.

The two collections are not registered equally well, and only the second should be relied on. The first is weak twice over. The registered document records its decision table as committed at 18:54:00Z on 2026-09-03, 76 seconds after the first model's `attribution` report was written — but that commit is not in the published history, since the instrument was split into its own crate the next morning and its history remade, so the hash resolves to nothing and that margin rests on my word. What anyone can check is the first commit on the public `preregistration` branch (<https://github.com/ljedrz/nachalnik/tree/preregistration>) at 20:38:11Z: sixteen minutes *after* the last of the first model's reports and eight minutes *before* the first request for the other five. So for one model of six the public record puts registration after the data. And a commit's timestamp is set by the machine that made it, not the server, so "public" means anyone can see the same timestamps I can, not that a third party vouches for them.

For the second collection, an amendment naming four thresholds — what would count as a replication, decided before there was anything to look at — went to that branch at 10:18:58Z on 2026-09-04, and the first request went out at 10:31:43Z. Thirteen minutes, both timestamps visible to anyone. That is the entire reason the re-run was worth \$15, and it is why the first collection's figures keep the weaker claim.

Re-reading the registered document against the record on 2026-09-17 turned up four errors in it, and one is a real deviation. The four: a power table rounded up in three of its four rows; two sections of the registration specifying different tests for H4, the own-versus-foreign hypothesis; H3 naming the wrong intervention, since the repair ladder never withheld the deletion it claimed to manipulate; and "the question's own arithmetic" being a residual category rather than a defined one. **None of the four touches the four replication thresholds or the §4.3 endpoint.** The power table changes 99% power to 98% at the registered size and the same design is chosen either way; H3's mislabelling leaves P5's numbers untouched and changes only what the ladder is said to measure (§4.5); the residual category leaves P2b met at 62 of 63 or 63 of 63 (§4.3).

The deviation proper is §8.2: a control-agreement gate that was registered and that the instrument never applied. It bears on the §4.3 endpoint, drops 35 items in the first collection and 33 in the second, and moves no verdict; §2.4 gives all three readings.

H4 is the one worth spelling out, because it is a duplicate test rather than an error of fact. §4 of the registration states P6 two-sided, per model — $|\text{own} - \text{foreign}| \leq 10 \text{ points}$ with an interval containing 0 — while §10 states the failure condition one-sided, as the own arm *beating* the foreign arm by more than 10 with an interval excluding 0, and is silent on the foreign arm leading. They come apart on this data: the two-sided band holds on four of six models and then three of six, while §10's failure condition is met by neither collection. **Neither reading was retired in favour of the other.** Both are reported separately in §4.4 and in the results documents, a decision taken on 2026-09-17 — after both collections, which is exactly why it was taken by reporting both rather than by choosing. A replication should fix the wording before collecting.

One registered rule did more than the rest, because it is what stopped me rescuing my own hypothesis:

┆ *No new hypotheses. If all of them fail, the honest paper is a negative one.*

All but one failed and no replacements were invented — §4.6 tests that rule rather than excepting it, since the most interesting thing measured here carries no prediction and is fenced off as a pilot. What the second collection established is narrower than a hypothesis and more useful than a null: the four thresholds held, so the effect the first collection found is not an artefact of the defect it was found with.

4. Results

Six models: `upstage/solar-pro4`, `deepseek/deepseek-v4-flash-0731`, `tencent/hy3`, `meituan/longcat-2.0`, `x-ai/grok-4.6`, `z-ai/glm-5.3-flash`. Six labs, and a range from a model that emits an answer with no visible deliberation to a frontier one. Each was measured twice: a first collection on instrument v4, run on 2026-09-03 and into the following morning, and a second on v5 on 2026-09-04, after the defect in §4.2 was found and removed.

The confirmatory claims are the four thresholds in §4.3, registered and timestamped before the second collection. Everything else here is exploratory or harness validation, is labelled as such where it appears, and is not used to extend those claims.

How to read the tables. A *dossier* — the harness calls it a *material* — is one planted context: nine *notes*, a three-way question, and one answer the notes support. An *item* is one note in one session, and it is what an ablation is run on. A *batch* is the set of copies asked under one condition, control or ablated. A *cell* is one dossier in one run of the repair ladder, and a *rung* is one stage of that ladder. An *arm* is one side of the own-versus-foreign comparison in §4.4. A *collection* is one sweep of all six models: the first on instrument v4, the second on v5. Throughout, denominators count the items actually measured, which is why they differ between models where a turn was cut, and numerators count claims that an item mattered.

4.1 Models are decent at saying what mattered

Each of the 54 items is one yes/no claim — *would removing this note change the answer?* — scored against what the copies did. The baseline is the best a subject with no self-knowledge could score by picking one answer and repeating it — not chance. Since most notes are inert, that constant answer is "no", so the baseline is the inert share of the items: `glm-5.3-flash`'s 76% is 41 inert notes of 54, the same 41 that §4.3 splits into 23 numeric and 18 plain. Both collections are exploratory here; no prediction covers accuracy.

model	first collection	baseline	second collection	baseline
glm-5.3-flash	52/54 — 96%	76%	46/54 — 85%	70%
hy3	48/54 — 89%	76%	46/54 — 85%	74%
grok-4.6	47/54 — 87%	69%	44/54 — 81%	69%
longcat-2.0	44/54 — 81%	81%	46/54 — 85%	81%
deepseek-v4-flash	41/52 — 79%	67%	39/53 — 74%	66%
solar-pro4	39/54 — 72%	80%	39/54 — 72%	74%

Four of six beat their baseline the first time and five of six the second — `longcat-2.0` moves from level with it to above it, and `solar-pro4` is the one model below it in both. This is the first place my expectations broke: any framing built on "models cannot report what their answers depend on" is contradicted by my own data.

Accuracy is the wrong single number, though, and the record carries the right two. Because most items are inert, a model can score well while missing most of what mattered. Recall is the share of load-bearing items it caught; precision is the share of its "this mattered" claims that were right.

model	recall, first	precision, first	recall, second	precision, second
glm-5.3-flash	13/13	13/15	11/16	11/14
hy3	11/13	11/15	11/14	11/16
grok-4.6	13/17	13/16	11/17	11/15
longcat-2.0	10/10	10/20	10/10	10/18
deepseek-v4-flash	10/17	10/14	10/18	10/15
solar-pro4	5/11	5/14	6/14	6/13
pooled	62/81 — 77%	62/94 — 66%	59/89 — 66%	59/91 — 65%

This reorders the models. `solar-pro4` and `deepseek-v4-flash` miss roughly half of what actually mattered — solar catches 5 of 11 and then 6 of 14 — so their 72–79% accuracy is mostly the credit for saying "no" to inert notes. `longcat-2.0` is the mirror image: it caught every load-bearing item in both collections, 10 of 10 twice, and paid for it with half its claims wrong, the second-worst precision in the cohort. `solar-pro4` is worst on both counts and in both collections, at 5 of 14 and 6 of 13.

How much of that spread is the subject, and how much is the sampling? Three copies decide each side, so a two-one plurality is what a coin flip looks like, and between 10 and 18 items per model were load-bearing where only six notes were designed to be (§6). Of the notes load-bearing for exactly one model, **15 of 20 in the first collection and 15 of 19 in the second rest on an arm that went two-one** rather than three copies against three. Restricting both arms to unanimous — the third rule of §2.4 — moves recall a long way and precision hardly at all:

reading	recall, first	precision, first	recall, second	precision, second
as published	62/81 — 77%	62/94 — 66%	59/89 — 66%	59/91 — 65%
§8.2 as registered	56/64 — 88%	56/82 — 68%	52/68 — 76%	52/77 — 68%
both arms unanimous	46/50 — 92%	46/67 — 69%	47/53 — 89%	47/66 — 71%

So most of the recall spread was stochasticity, and one part of it was not. Where both arms agreed with themselves, five of the six models catch 80–100% of what mattered; `solar-pro4` is the exception at 4 of 6 and then 3 of 7, the only one below 80% and worst on precision under every rule. So the precision finding stands, and **the recall ranking should be read as provisional**, because a third of the determinations behind it rest on a single copy having moved. More copies per arm is the cheapest useful thing anyone could do to this design.

4.2 An item's number was not a fair question

The same models were asked, about the same notes: *what number is this note in your context?* Eighteen questions each. `solar-pro4` got two right and the other five got none.

2 of 108 — and it is not a result at all. An earlier draft called it the study's cleanest finding; it was withdrawn on 2026-09-04 and v5 removed the probe. The projector renders each item as `label:` and its content, never a number, and `attribution` installs no handles, so these subjects had no way to see the numbering they were being asked about: they were asked for a value they could not observe. The score says so. 1.9% against a 33% majority baseline is an order of magnitude *below* what a constant guesser gets, which is the signature of an unanswerable question rather than an absent faculty (§5, point 1). `nachalnik` *can* address items to a model — `inspect`'s `look` lists every one by number — and that handle simply was not granted here, so **this says nothing about whether a model can locate an item when allowed to look**. That question is open.

4.3 The errors have a shape

Looking only at notes classified as inert under the scoring rule of §2.4 — control and ablated batches landing on the same unique plurality answer:

`red herrings` and `other numeric partition numeric claimed` rather than adding to it, so a row is `numeric + plain` inert notes and the rest of that model's items are load-bearing: `longcat-2.0`'s first row is 26 + 18 = 44 of 54, which is its §4.1 baseline of 81%.

First collection, instrument v4 — exploratory. The hypotheses were registered, but the defect in §4.2 was still in the instrument when these were taken.

model	numeric claimed	plain claimed	red herrings	other numeric
longcat-2.0	10/26	0/18	0/12	10/14
solar-pro4	9/27	0/16	0/12	9/15
deepseek-v4-flash	4/19	0/16	0/10	4/9
hy3	4/24	0/17	0/12	4/12
grok-4.6	3/21	0/16	0/11	3/10
glm-5.3-flash	2/23	0/18	0/12	2/11
total	32/140	0/101	0/69	32/71

"Other numeric" is the residual: every inert note carrying two adjacent digits that is not on its dossier's planted-decoy list. It is the question's own arithmetic in all but one case, named below; the denominators count every such note measured, over-claimed or not.

Second collection, instrument v5 — confirmatory. Same six models, same material, location probe removed. The four thresholds these are read against were committed to a public branch thirteen minutes before the first request.

model	numeric claimed	plain claimed	red herrings	other numeric
longcat-2.0	8/26	0/18	0/12	8/14
solar-pro4	7/24	0/16	0/12	7/12
deepseek-v4-flash	4/19	1/16	0/10	4/9
hy3	5/23	0/17	0/11	5/12
grok-4.6	4/21	0/16	0/10	4/11
glm-5.3-flash	3/23	0/15	0/11	3/12
total	31/136	1/98	0/66	31/70

Three counts here, and a caveat on the third. None of them is a rate, and all three span both collections.

Not one model, in either collection, claimed a red herring mattered. 0 of 135. These six models are not fooled by irrelevant numbers.

One note without figures was claimed, out of 199. The first collection had none at all; `deepseek-v4-flash` claimed one in the second, which is why this sentence no longer says "never".

62 of the 63 false positives carrying a figure were notes belonging to the question's own arithmetic that happened not to decide it. Every model showed it in both collections, and the effect ran the same way for all six each time (sign test; Appendix A gives $p = 0.031$ two-sided for the first collection and $p = 0.016$ one-sided for the second, where the direction was registered in advance).

The remaining one is worth naming, because the category is looser than its name. The instrument sorts a numeric inert note into "red herring" if it is on the dossier's planted-decoy list and into "the question's arithmetic" otherwise, and "numeric" means any two adjacent digits anywhere in the text. `mill/records/guidance` — "Assessors read the Luza column first: Luza has been the binding constraint in every quarter since 2019" — trips

that test on the year and carries no quantity at all. `longcat-2.0` claimed it once, in the first collection. The other 62 sit on a capacity, an intake, a rate, a stock or a threshold the question really is computed from, so the finding survives. But the accurate name for the category is *notes central to the task that did not decide it*: "arithmetic" describes 62 of the 63 rather than defining the bucket. Counting the plain note above, that is 64 endpoint false positives in total.

My registered hypothesis was that models over-weight *digits*. That is wrong. They over-weight *things that look like the reasoning*, which is a different and much more sensible failure. I registered that distinction in advance as the outcome most likely to be mistaken for the other one, which is the only reason I am able to say it cleanly now.

Where the errors land is a count and stands. *Why* they land there is a reading, and a note that belongs to the question's arithmetic differs from an inert one in more than topic — length and position travel with it, and this material was not built to pull those apart.

4.4 Reading your own context looks like just reading

No own-context advantage was detected. Each model was asked the same kind of counterfactual question twice: once about its own context, and once about a transcript of a different session's context, quoted to it. Five items per arm, less any turn that was cut — which is why `deepseek-v4-flash` has a denominator of 4 in the first collection.

model	own, first	foreign, first	own, second	foreign, second
solar-pro4	1/5	3/5	1/5	3/5
deepseek-v4-flash	1/5	1/4	3/5	3/5
hy3	3/5	3/5	4/5	3/5
longcat-2.0	3/5	3/5	3/5	5/5
grok-4.6	3/5	2/5	3/5	3/5
glm-5.3-flash	4/5	4/5	5/5	5/5
pooled	15/30	16/29	19/30	22/30

Reflecting on its own context bought a model nothing: three exact ties each time, and pooled it came out slightly worse both times. Nor are the two arms interchangeable per model. The own-minus-foreign difference falls within the registered ten points on four of six models the first time and three of six the second, and where it misses it misses by twenty to forty points, in both directions.

Those are the two registered readings, and the registration states them differently (§3). The ten-point band above is P6 as §4 of the preregistration puts it, two-sided and per model. §10 states the failure condition one-sided instead — H4 fails if the *own* arm beats the foreign arm by more than ten points with an interval excluding zero — and **neither collection meets it**, since every per-model interval contains zero: the two models whose own arm led by more than ten points have intervals running −32 to +60 and −31 to +60. Both readings are reported because neither was retired in favour of the other after the data was in.

What holds without exception is the weaker thing: every per-model interval contains zero, both times, as does each pooled one. At five items per arm per model this is a failure to detect. It is not the sign of an effect, not equivalence between the arms, and not evidence that privileged access is impossible.

4.5 Telling beats measuring

The last experiment plants a false note that contradicts the records, then asks the same question five times, adding exactly one thing between each pair:

1. **carrying** — it just has the false note
2. **again** — the same question, a second time, nothing added (a control)
3. **unprompted** — it is given tools to inspect and edit its own context, with no hint anything is wrong
4. **told-so** — it is told one of its notes contradicts the records, and names which
5. **repaired** — it is asked to fix it

Five of the six dossiers (the ladder does not use `mill`) with three independent ladders each: fifteen observations per rung per model, less the odd cell a truncated turn took out. Every rung is scored over all fifteen cells, the ones the falsehood never took included, which is why `carrying` reads as the share of cells the lie failed to fool. The contrasts quoted after the table are paired over the cells where both rungs were measured, as registered, so where a turn was cut they are not differences of these columns. Each cell is first collection, then second:

model	carrying	again	unprompted	told-so	repaired	used the tools
solar-pro4	5/15, 6/15	5/15, 6/15	0/15, 0/15	1/15, 3/15	2/15, 0/15	27%, 25%
longcat-2.0	3/15, 3/15	3/15, 2/15	2/15, 2/15	7/15, 3/14	2/15, 3/15	37%, 37%
deepseek-v4-flash	2/15, 3/15	1/15, 1/15	5/15, 1/15	8/15, 7/14	10/15, 8/14	50%, 47%
hy3	3/15, 6/15	6/15, 7/15	6/14, 11/15	14/15, 13/15	11/15, 13/15	62%, 62%
grok-4.6	4/15, 5/15	3/15, 5/15	4/15, 5/15	9/15, 9/15	9/15, 9/15	67%, 68%
glm-5.3-flash	8/15, 9/15	6/15, 7/15	9/15, 8/15	10/15, 10/15	10/14, 14/15	60%, 58%

Models using the tools on fewer than half the questions they could have were excluded from this comparison in advance: two of six the first time, and three the second, because `deepseek-v4-flash` came in at 47% having cleared the same bar at exactly 50%. The rule is pre-specified and was applied as written rather than nudged — see §5, point 10. `solar-pro4` is worth a sentence on its own: in the first collection it made 16 edits off 6 tests, rewriting its own context nearly three times as often as it measured anything, and 13 off 7 in the second; both times its answers got *worse* with tools than without them (5/15 down to 0/15, then 6/15 down to 0/15). That is a small, concrete warning about giving edit authority to weak models.

Among those that cleared the bar — four models the first time, three the second — here are the three registered contrasts in percentage points, paired over the cells where both rungs were measured:

model	tools, no hint	told	asked to fix
<i>first collection</i>			
deepseek-v4-flash	+27	+20	+13
glm-5.3-flash	+20	+7	+7
grok-4.6	+7	+33	0
hy3	+7	+50	-20
<i>second collection</i>			
glm-5.3-flash	+7	+13	+27
hy3	+27	+13	0
grok-4.6	0	+27	0

Being told helped every model, both times. Four out of four, then three out of three.

Being handed tools with no hint helped more than registered. P3 put the ceiling at five points; every model in the first collection cleared it and two of three in the second, though three of those six gains are seven points and two do not survive the reading below.

Being asked to put the context right moved one measurement of seven. glm-5.3-flash went 10/15 to 14/15 in the second collection, one model went twenty points the other way, and the rest sat between. At $n = 15$ none of these is a precise estimate and the registered one-sided ceiling is not an equivalence test, so what the ladder supports is that the instruction did not reliably help, not that its effect is zero.

Restricting the reading to the cells the lie actually fooled changes no direction. The tables above score every cell, including the ones where the falsehood never took (§2.4). The lie took hold in 43 of 60 cells in the first collection and 25 of 45 in the second, among the models that cleared the bar. Over those cells the dilution comes off and the pooled contrasts sharpen:

contrast	first collection	second collection
tools, no hint	+14	+16
told	+37 (17 gained, 1 lost)	+32 (8 gained, 0 lost)
asked to fix	-2	+12

The two small seven-point gains from the tools disappear under this reading — grok-4.6's in the first collection and glm-5.3-flash's in the second — because the single cell each came from was a cell the lie had never fooled. cargo run --release prints both readings.

The registered prediction was that the deletion would be what mattered, and this ladder cannot say whether it does, because it never withheld deletion. The tools arrive at unprompted and are never taken away, so told-so and repaired differ in what the subject is told, not in what it may do; the last rung adds "Something in your context is wrong. Find out what, and put it right." That is deliberate — a rung that also handed over a new capability would confound the two — but it means no sentence here measures the value of granting edit authority. What the ladder can say is that disclosure moved the answer and the instruction to act on it mostly did not, for the next answer only. The subjects bear it out: 10 edits across both collections over the two rungs where they held the

handles unasked, and 228 once they were told to put the context right. The rungs always run in this order and the later ones have had more turns in which to edit, so that ratio describes the ladder as run rather than a controlled comparison.

4.6 A capability, measured and not predicted

No prediction covers anything in this section. It was registered as harness validation — a check that the ablation handle returns what it should — and that is the only claim it supports. It is reported because leaving it out would be worse, and it is fenced because the last figure this paper reported without a prediction attached turned out to be an artefact (§4.2).

`instrumented` asks the same counterfactual three ways: of a subject with no handle, of that same subject once given one *after* it has already answered, and of a fresh subject given one that was never asked to guess.

model	reported	retested	tested	reported → retested	reported → tested
glm-5.3-flash	45/53	44/53	48/54	-2 (p = 0.73)	+4 (p = 0.38)
hy3	43/54	45/54	49/54	+4 (p = 0.36)	+11 (p = 0.09)
grok-4.6	36/54	39/54	47/54	+6 (p = 0.29)	+20 (p = 0.013)

Every model used the handle on essentially every question — 98%, 100%, 100%, and 353 tests between them — so whatever this is, it is not a failure to reach for the tool.

Three of three point the same way on `reported` → `tested`. On `reported` → `retested` it is two up and one down, none of the three significant, and the first collection's single `glm-5.3-flash` run — the stage-1 harness check — had the same shape: level on `retested`, up on `tested`. But **tested differs from reported in two ways at once**: it has the instrument *and* it never committed to an answer first. Nothing here can say which of those is doing the work, and the obvious fourth arm — a subject that has not spoken and has no instrument — does not exist in this design.

So this is a pilot for an experiment, not an experiment. It is written down here with its confound named so that the registered version has something to be compared against.

5. What it takes to measure this honestly

Forking a session and removing an item is a weekend's work. Everything below is what the rest of the time went on, across two collections, and each line is something I got wrong at least once.

1. Baseline against the best constant answer, not against chance. This is the one that caught me. The location probe scored 2 of 108 against a *majority* baseline of 33%. Landing an order of magnitude under the do-nothing baseline is not a weak faculty, it is a broken question, and the harness had been printing that number all along.

2. Test your preconditions and print them. In the second collection the harness flagged 32 of the 90 repair-ladder cells because the planted falsehood never fooled the subject in the first place. The flag is what stops a model too sharp to be fooled from reading as a model that failed to repair; §4.5 gives the ladder both ways.

3. Restrict the endpoint to items where the honest answer is known. The primary measurement looks only at notes whose removal did not change the scored answer, because there the correct claim is "no" for every item. Across the material as a whole, notes with figures really are more likely to matter — an unrestricted comparison would pay a model for a lucky prior.

4. Ask about two copies, not about the model. "Would *your* answer change" is a different question from "will these two copies differ", and a subject can be right about the copies and scored wrong because the live session — which has the elicitation in its context — answered unlike both. I learnt this from a model that was right while I was marking it wrong.

5. Blind the copies to the answer already given.

6. Separate your failure from the subject's. A turn that ran out of tokens having said nothing is your ceiling, not a wrong claim, and is excluded. An unreadable answer from a subject that *did* speak is the subject's failure and is scored as one.

7. Fingerprint the questions. Otherwise "we re-ran it" is an assertion. When the location probe came out, two of seven fingerprints moved and five did not, and the record says which.

8. Grant looking and changing separately. A tool that answers "may it experiment on itself" must not also answer "may it rewrite its own memory". One model in this cohort rewrote its own context nearly three times as often as it measured anything, and got worse for it (§4.5).

9. Use one frozen origin for both measurements. When a subject tests its own context, it should fork from the same snapshot the harness scores against — otherwise its evidence and your ground truth are two different experiments.

10. Apply your gates as written, even when they cost you. The handle-use gate excluded a third model in v5 because it came in at 47% against a pre-specified 50%, having been at exactly 50% the first time. Three points, one model, no adjustment.

11. Correct for clustering. Nine items from one dossier are not nine independent observations.

12. Write down what would count as a replication before you collect it. The only real degree of freedom in a re-run is what counts as the same answer, and without it fixed in advance, close numbers read as replication and divergent ones read as explanation. Both feel honest at the time.

5.1 What the measurements say

Secondary to the above and shorter-lived, one line each, with the section that argues them.

- **Don't score this with accuracy alone.** Most of a context is inert, so "no" to everything already scores 66–81% here; and check any recall spread against your replicate count, because most of this one was which copies flipped (§4.1).
- **Expect false positives in a specific place** — not on planted irrelevant numbers, but on material that is topically central and causally inert (§4.3).
- **Don't assume self-inspection is privileged.** No own-context advantage was detected, which at five items per arm is a failure to detect rather than a demonstration that none exists (§4.4).
- **Tell your agents things.** The *information* that a note is false carried more than the instruction to act on it — but the ladder scores the answer a subject is about to give, not what its context carries afterwards, and I stated that too broadly first (§4.5).
- **If you ask a model to name an item, give it a way to look first.** With no handle, every model answered confidently and wrongly rather than declining; whether they manage it *with* one is untested here and is the obvious next experiment (§4.2).

6. Limitations

I would rather write these than have them found.

Six models, one API, two collections, one pair of hands. The second collection is a replication in the sense that matters — fresh data against thresholds fixed beforehand — and in no other: same six models, same provider, same author, a day apart. Provider routing was not controlled. Nobody independent has run this.

The materials are synthetic and all of one shape — three options, an arithmetic question, a buried correction. Whether any of this generalises to real agent contexts, with tool output and long histories, is unknown and not claimed.

The planted decisive notes were not reliably decisive. Each dossier was built around one note that should have decided the answer. Identically in both collections, one of the six moved all six models (`depot/records/omsk-annex`), three moved five, one moved four, and `mill/records/guidance` moved one — the last by design, `mill` being built so that what the notes *support* and what a model *uses* come apart. The rest is the ordinary failure: the material specifies what *should* be load-bearing and the subject decides what *is*. Only two notes in the set moved every model, and one is not a planted pivot at all but `mill/records/consumption`, that dossier's denominator. So ground truth cannot be assumed fixed across models, and at three copies per arm the design cannot always tell an idiosyncratic dependence from a coin flip (§4.1).

The foreign arm arrives quoted where the subject's own arrives as context. That is the difference §4.4 tests and also, unavoidably, a difference in presentation. The quotation is rendered exactly as the projector renders the subject's own notes, which is the mitigation; the registered counterbalance — running the two dossiers the other way round — was never run, in either collection. The null in §4.4 is read with that confound unremoved.

The hypotheses came from exploratory data. I found them by looking at early runs, then registered them and collected fresh data. Standard practice, but it means the exploratory runs cannot also count as evidence, and they do not.

In the first collection the withdrawn location probe ran first, so every v4 claim in §4.1 and §4.3 was made in a context where the subject had just produced three confident, wrong item numbers. v5 removed it and re-ran everything — the pooled difference moved from +23 to +22 and every registered threshold held — so the concern was tested rather than argued away, but the first collection's absolute rates could still be moved by it.

The ablation handle is confirmed on three model families, not six. The check ran on one model in the first collection and three in the second (§4.6). The other three were never checked this way.

7. How this was made

Given the subject matter, this section is not optional.

The great majority of this work was done by a language model (Claude Opus 5), working interactively with me over one long session. It wrote the runtime harness, the experiment code, the statistics, the preregistration, the analysis tooling, and the first draft of this paper. I set the direction, made every judgement call about scope and spending, funded it, and pushed back — several times decisively, including on the framing of related work and on the choice of models.

What makes it checkable regardless of who wrote it is that **you do not have to trust the process:**

- the preregistration is on a timestamped public branch, dated before most of the data existed;

- every question put to every model and every answer received is saved verbatim;
- every number in this paper recomputes from those records, and the primary endpoint, the four replication thresholds and both readings of the repair ladder come out of one command;
- Appendix B logs what changed after the freeze, and PREREGISTRATION.md §11 has all twenty-one dated entries, including the unflattering ones.

8. Related work

Apparatus. Ground truth about causes requires intervention rather than observation (Pearl, 2009), and that intervention has reached language models at every layer, this one only lately: causal mediation analysis (Vig et al., 2020), interchange interventions (Geiger et al., 2021) and activation patching (Meng et al., 2022) all work on internal representations. At the prompt level, ERASER (DeYoung et al., 2020) erases the tokens a rationale names and measures what changes, which is this measurement on a static string. What a runtime adds is that the string is a live session, so the copy resumes from the same frozen point with everything but the named item identical.

The nearest neighbour is Causal Agent Replay (Shah, 2026), which treats a run as a structural causal model, applies `do(·)` to a step and re-executes forward under the same policy — the manoeuvre every claim here is scored against. It intervenes on *steps* rather than named context items and uses no self-report at all, validating against planted ground truth, so the comparison this paper is built on — what a model says against what a fork does — is not one it attempts.

VISTA (Xu et al., 2026) reaches the same diagnosis from the other end. Its first contribution is that frontier models are "proprioceptively blind to their own context", and its answer is a dashboard of per-block metadata with archive and recover tools. That is concurrent work, arrived at independently and found during the literature review rather than built on, and the variable it makes visible is not this one: VISTA's is *magnitude* — block size, recency, remaining budget — where this is *causal dependence*. The distinction is one of kind rather than degree. No amount of metadata display answers "would I have said something else without note 4", because a counterfactual has to be run. VISTA makes context legible; this makes it testable.

Treating context as addressable memory is otherwise well established, for other purposes: MemGPT (Packer et al., 2023) pages it like virtual memory, LLMLingua (Jiang et al., 2023) compresses it, Active Context Compression (Verma, 2026) consolidates a trajectory and deletes the raw logs, Self-GC (Hao et al., 2026) prunes indexed context objects by predicted usefulness rather than by provenance, and Slipstream (Chen et al., 2026a) validates a compaction summary against the agent's independently continued reasoning. The premise is shared and the purpose is not: the same primitives verify a claim here rather than fit a budget.

Findings. Self-report about context is already benchmarked, and the accuracy of such reports is not a new question. Zeng et al. (2026) put nine kinds of question to models about their own behaviour — whether a prompt edit would change the answer is one of them — check the answers by resampling, find the skill real but limited, and then train models to be better at it. Naphade et al. (2026) run a third-party control and report a small significant self-advantage. This is not a new benchmark of that, and does not claim to be. One part of Zeng et al. is closer still. To make training data, their pipeline rewinds a conversation to an earlier turn, changes one thing, replays it, and has a judge score what the model then does. That is the move this apparatus makes, and it is published first. The purpose is the opposite: there the fork manufactures examples and a model is trained on them; here nothing is trained, and the fork is the measurement. What is asked differs too. Their model predicts a judge's score for an edit it is told about; here the subject is asked about the context it is sitting in, and the copies that check it have one item removed rather than rewritten.

That post-hoc explanations are frequently unfaithful is settled too (Jain and Wallace, 2019; Jacovi and Goldberg, 2020; Atanasova et al., 2023): Turpin et al. (2023) show plausible rationales that never mention the feature actually steering the model, and Lanham et al. (2023) show that perturbing chain-of-thought steps sometimes leaves the decision unmoved. §4.3 sharpens the shape rather than the fact — the false attributions go not to digits as such but to notes belonging to the question's own arithmetic that happened not to decide it. What gets over-claimed is not what looks numerical but what looks like a reasoning step.

Self-correction and repair. Huang et al. (2023) find that without external feedback models struggle to identify and correct their own errors, and that performance sometimes degrades. §4.5's ladder is that finding with the content of the feedback varied: disclosure repaired the answer for every model that cleared the instrumentation bar, while the instruction to act on it cleared the registered ceiling in one measurement of seven. The nearest published work is MemSecBench (Chen et al., 2026b), which measures whether an agent, told only to audit itself, removes poisoned long-term memory and keeps the benign kind. It measured that first, in the security framing, and two things differ. Its repair stage is scored on the state the memory ends up in, and what the poison went on to do is scored separately, at an earlier stage; this ladder scores the next answer to the task. And its agent is given no hint about which item is at fault and is never asked to name one, where here the subject names the item on the record before it is *asked* to change anything — it could already, having held the handles since unprompted — which is what makes naming and fixing separable at all, and lets the answer after being told and naming be compared with the answer after being asked to put the context right.

Introspection and causal bypassing. Binder et al. (2024) find that a model finetuned to predict its own behaviour beats a differently-trained model at it, and read that as privileged access; Lindsey (2026) injects concept vectors and finds that models can notice, the strongest positive result in the field; Singh et al. (2026) is a recent reality check on that literature. Neither cohort overlaps this one — Binder et al. work on GPT-4, GPT-4o and Llama-3, Lindsey on Claude models — and both required finetuning or white-box access where these measurements are zero-shot and black-box. §4.4 also asks a different question, not *what will I say* but *what is my answer made of*, and finds no own-context advantage. The two are compatible only if self-knowledge is not one faculty.

That null is what causal bypassing predicts (Morris and Plunkett, 2025): a report can be accurate by a route that never passes through the state it reports, and a subject here can get an ablation right by reasoning about the task alone. Predicts is not shows. §4.4 reports the absence of an advantage and not its sign, at five items per arm per model and with the foreign context arriving quoted (§6). These numbers do not require privileged access to explain them; at this size they also do not rule it out, and they do not establish what the model is doing instead.

This section was assembled by a language model. Three of its characterisations have since been checked by the works' own authors: Zeng et al. (2026), whose first author pointed me to the multi-turn track described above; VISTA, whose first author asked for no change; and MemSecBench, whose corresponding author asked that the scope of its repair stage be stated more exactly, which it now is. The rest has not been checked by anyone who reads this literature regularly. It is the one part of the paper I cannot verify from my own data, and I would welcome corrections.

What I can now say is narrower. Every identifier in §10 was resolved against arXiv on 2026-09-06 and matches the title and authors recorded there, so the works exist and their abstracts say what this section says they say. That is not the same as having read them closely enough to have placed them correctly, which is what the paragraph above is about.

9. Reproducing this

This study is its own repository, <https://github.com/ljedrz/deleting-a-memory>, holding the saved runs, the analysis and the write-ups. The runtime and the instrument are a separate one, `nachalnik` at <https://github.com/ljedrz/nachalnik>, pinned here at commit `d3b3ba6` — deliberately, so that a second study starts from a harness rather than a copy of this one, and so that the instrument cannot learn what this study registered.

```
cargo run --release
```

reads `eval-runs/` and recomputes §4.3's tables for both collections, §4.1's recall and precision under each of the three rules, the count of notes load-bearing for exactly one model split by whether either arm went two-one, and the repair ladder over the cells the falsehood actually fooled. It reports each of the four replication thresholds as a verdict rather than a figure, because what counts as a replication was fixed in advance and printing only the number would hand that reading back to whoever is looking. Nothing is re-requested and nothing costs anything; the collections themselves cost \$9.11 and \$15.09.

`PREREGISTRATION.md` holds what was fixed in advance, its §11 the dated log of every deviation; `RESULTS.md` and `RESULTS-v5.md` map each registered prediction to its outcome; `METHODOLOGY.md` is the full methods document, with the eleven known threats to validity, three of them uncontrolled; `RUNBOOK.md` is what was run, in what order, and what it cost.

10. References

Every arXiv identifier below was resolved against arXiv on 2026-09-06; the titles and authors are as recorded there. `RELATED.md` holds the longer positioning notes, including work this paper does not cite, and was checked on 2026-09-03.

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Appendix A: the arithmetic, by hand

The main claims are counts, and can be checked without any statistics at all.

The endpoint counts. Both collections, as printed in §4.3: red herrings 0 of 69 and 0 of 66, 0 of 135 in all; plain inert notes 0 of 101 and 1 of 98, the one claimed by `deepseek-v4-flash`; numeric over-claims 32 and 31, all 63 of them off the red-herring set, which is the whole of the discrimination result. Of the 63, **62 are notes belonging to the question's own arithmetic and one is not:** `mill/records/guidance` carries no quantity and is filed there only because its text contains a year (§4.3). Under the §8.2 gate the same counts are 26 and 25, with the same single exception among them.

Location. 18 questions per model, 6 models = 108. Correct: 2, both from `solar-pro4`. §4.2 explains why this does not measure what it looks like it measures.

Recall and precision (§4.1), and a trap in deriving them. Each model's 54 claims sort into four cells: caught a load-bearing item, missed one, claimed an inert one, correctly dismissed one. Recall is the first over the first two;

precision is the first over the first and third. Pooled, that is 62/81 and 62/94 in the first collection, 59/89 and 59/91 in the second.

It is tempting to derive recall from §4.1 and §4.3 instead — accuracy minus the inert items answered correctly — and it is right in eleven of the twelve cells. It fails on `deepseek-v4-flash` in the second collection, where it gives 9 of 18 against a true 10 of 18. The reason is that §4.3's counts are *claims that an item mattered*, and one of that model's items was scored wrong without being one of them: the subject spoke and its answer could not be read, which is scored as the subject's failure (§5, point 6). So errors on inert items exceed §4.3's false positives by exactly one there. The four cells are read straight off the record instead.

Sign test on direction. Six models, all six with the same sign, under a null of "either direction equally likely": $(1/2)^6 = 1/64 = 0.0156$. That is the one-sided figure, and the right one for the second collection, where the direction was registered in advance. For the first, where it was not, the two-sided figure is $2/64 = 0.031$.

Confidence intervals in the repository's output are Wilson score intervals. The accuracy scores are widened for clustering by dossier where more than one dossier contributes, with the unadjusted interval printed alongside so the cost of the adjustment is visible. **The primary endpoint's intervals are not:** `Surface` is computed from four counts and is handed no cluster information at all, so its numeric and plain intervals are plain Wilson intervals over items that are in fact clustered by dossier, and are correspondingly too narrow. No claim in §4.3 or §5.1 depends on one — they are counts — but the intervals printed there should not be read as adjusted.

Appendix B: what changed after the freeze

One line each, dated. `PREREGISTRATION.md` §11 is the authoritative version, with all twenty-one entries and their reasoning; where the main text carries the fact, the section is named here.

- **2026-09-03, before any confirmatory run.** The shared brief told every subject it had no tools, which was false of the two experiments that hand over handles; a second brief was added for those two, moving their digests and no others.
- **2026-09-03, before any confirmatory run.** A primary endpoint turned out to be circular — the tool printed the answer it was being scored on — and was replaced.
- **2026-09-03, before any confirmatory run.** The material set could not falsify its own hypothesis, so red herrings were added; instrument v4 breaks comparability with v3 to do it (§2.2).
- **2026-09-03.** Two models were substituted, neither having produced a figure bearing on any hypothesis: one could not complete a run through its provider, which cut it off after about fifteen requests every time; one was swapped for a stronger and cheaper alternative in the same slot.
- **2026-09-03, and the first collection's whole registration.** The commit the registered document cites for its decision table is not in the published history, and the public branch postdates one model's results (§3).
- **2026-09-04 — a result withdrawn.** The 2-of-108 location figure, an earlier draft's cleanest finding, is an artefact of asking for a number the subject could not see; v5 removed the probe (§4.2). `tests/machinery.rs` now requires that any experiment asking for an item's number also grant a handle to look at the numbering, and pins the two frozen experiments that do not.
- **2026-09-04.** The registered commitment to run the handle check on three models was not kept on the day — the first attempt died on an exhausted key limit — and was kept later the same day.
- **2026-09-04.** The pre-specified handle-use gate excluded three models rather than two, because `deepseek-v4-flash` came in at 47% against a 50% bar it had cleared exactly the first time. Applied as written (§5, point

10).

- **2026-09-06.** A commit the preregistration cites no longer resolves, for the same history-rewrite reason as the entry above.
- **2026-09-17, after both collections.** Four errors found in the registered document itself while re-checking it against the record: a power table wrong in three of its four rows, two sections registering different tests for the same hypothesis, §8.2 registered as a gate the instrument never applied, and H3 naming the wrong intervention (§3). Also that "the question's own arithmetic" is a residual category rather than a defined one (§4.3).
- **After drafting — a second result narrowed.** A draft read the ladder as showing that deleting a false note was worth nothing; it weakened when `glm-5.3-flash` went 10/15 to 14/15 on the last rung, and was wrong in kind too (§4.5).
- **Undated, found in the analysis.** A script silently dropped a model from the results. Fixed; the command now names any report it could not parse rather than skipping it in silence.